28 June 1974
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

MORTAR, 60MM: M19

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of high-angle fire mortar cannon M19 with mount MS and baseplate.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-1 3931	Cannons, General Specification for
MIL-P-14232	Parts, Equipment and Tools for Army Material, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage

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PUBLICATIONS

U.S. Army Armament Command

EPL 7238469 - Engineering Parts List, Mount; 60MM Mortar: M5
EPL 7238503 - Engineering parts List, Cannon, 60MM Mortar: M19
DL 7229669 - List of Gages, 60MM Mortar: M19
DL 7238469 - Index of Inspection Equipment, Mount Assembly; 60MM Mortar: M5
DL 7238503 - Index of Inspection Equipment, Cannon Assembly; 60MM Mortar: M19
LSQAP 6536240 - List of SQAPS, Baseplate Assembly; 60MM Mortar
MLSQAP 7238469 - Master List of SQAPS, Mount Assembly; 60MM Mortar: M19
MLSQAP 7238503 - Master List of SQAPS, Cannon Assembly, 60MM Mortar: M19
PDS P6732010 - Packaging Data Sheet; 60MM Mortar: M19

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 First article. When specified in the contract, the contractor shall submit a first article. The article shall be specified in the contract. For example, Cannon, Mount, Baseplate or any part or portion thereof. Unless otherwise specified (see 6.2.1), the first article shall be included in the first article package (see 5.1).

3.2 Materials, design and construction. The cannon and the mount shall conform to the drawings listed in Engineering Parts List EPL 7238503 (Cannon Assembly), EPL 7238469 (Mount Assembly), to this specification, and to MIL-C-13931.

3.3 Material soundness. Material of the tube, cap, housing, firing mechanism, mount (bipod), and baseplate shall be free from cracks and shall comply with the requirements of the applicable drawings and specifications.

3.4 Assembly of mating parts. Assembly of the breech face of the tube to the contact shoulder in the cap, the firing pin bushing to the adapter and the adapter to the housing shall be as specified on the applicable drawings Assembly of parts shall also conform to the requirements of MIL-C-13931.

3.4.1 Firing pin protrusion. Protrusion of the firing pin from the face of the base plug bushing shall be as follows:

a. Drop-fire: 0.047 to 0.101 inch, incl.
b. Lever-fire: 0.085 to 0.101 inch, incl.

3.5 Functioning after assembly

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3.5.1 Elevating mechanism. The tube with bracket for the connector assembly, and clevis pivot. The ring retains all of the moving parts within the tube. All parts shall be aligned as specified on the applicable drawings. The elevating mechanism shall function through the entire operating range. Backlash shall not exceed 1/8 turn. (D 7228725)

3.5.2 Traversing mechanism. The bearing (B5025227) on the shouldered end of the screw-traversing (B7228286) shall be adjusted to obtain a smooth continuously sliding fit. The bearing can be adjusted to only one location in the tube. Geometry of the parts determine the fit at other locations. Backlash shall not exceed 1/8 turn.

3.5.3 Cant mechanism. Leg, Left, Assembly (D7230150). The mechanism incorporated into this assembly shall provide a smooth, positive adjustment throughout the range. The coil spring and washers in the nut cant adjusting assembly increase the firmness of the joint and the stability of the adjustment during firing. The bracket when not locked on the sleeve shall move easily for preliminary leveling and shall grip the sleeve securely when the sleeve (B7230007) is tightened manually. The sleeve (A5025237) shall move smoothly on the leg body.

3.5.3.1 Cant mechanism adjustment. The nut, cant adjusting is connected to the sleeve (A5025237) by the ring (A5025233). The ring shall be screwed onto the nut, cant adjusting until the end of the nut is seated on the end of the sleeve. The ring shall then be unscrewed the minimum distance necessary to permit manual rotation of the nut while the sleeve is locked against rotation. Following this adjustment, the ring and the nut, cant adjusting shall be drilled and tapped to receive the screw (A5025235).

3.6 Performance. The mortar assembly (cannon and mount) shall be capable of withstanding the stresses of firing and functional operation under service conditions. The cannon shall be capable of withstanding a chamber pressure of 6800 ± 200 psi at 70°F as specified in 4.6.5.

3.6.1 Functioning. There shall be no impairment in the functioning of the mortar assembly (cannon and mount) during and after the testing specified in 4.6.3, 4.6.4, and 4.6.5.

3.6.2 Bore enlargement. Bore diameters (from the muzzle to a depth of 24 inches) after proof testing shall not exceed the basic [minimum] diameter specified on the drawings by more than 0.004 inch. There shall be no indication of abnormal wear, damage or deformation of the bore surface.

3.6.3 Material soundness. After completion of testing, all parts shall be free from cracks and fractures and shall comply with soundness requirements specified on the applicable drawings.

3.7 Operation. All parts shall operate without interference, erratic movement or malfunction.

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3.8 Finish.

3.8.1 Surface texture. Surface texture shall be as specified in MIL-C-13931.

3.8.2 Painting. Where painting is specified for phosphate finished parts, application of supplementary oil shall be omitted.

3.9 Marking. Marking of parts shall be as specified on the drawings and in MIL-C-13931.

3.10 Workmanship. Workmanship shall be as specified in MIL-C-13931.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used in this specification are in accordance with MIL-STD-109.

4.3 Lots. The size of lots of component parts shall be as approved by the Government representative, subject to the provisions of MIL-C-13931.

4.4 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.5)
b. Quality conformance inspection (see 4.6)

4.5 First article inspection. A first article shall be submitted for inspection in accordance with the contract requirements. The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspections specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.6 Quality conformance inspection.

4.6.1 Inspection provisions. Mortars shall be inspected for conformance to the requirements of the provisions of this specification, to MIL-C-13931, to the SQAPS listed in LSQAP 6536240, MLSQAP 7238469, LSQAP 7238503 and to SQAP-APPENDIX-WVT.

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4.6.1.1 Final examination. Final examination of each mortar assembly (cannon and mount) shall be completed after testing and prior to preservation and packaging to determine compliance with this specification, MIL-C-13931, and as specified in the contract.

4.6.1.2 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

4.6.2 Inspection equipment. Data lists containing drawing numbers of Government designed inspection equipment used in the inspection of parts, subassemblies and assemblies are listed in DL 7238503 and DL 7238469.

4.6.2.1 Acquisition, maintenance and disposition. Unless otherwise specified, (see 6.2.1) responsibility for acquisition, maintenance and disposition of acceptance inspection and test equipment prescribed on the List of Inspection Equipment DL 7238503 and DL 7238469 and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.6.2.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.6.3 Examinations.

4.6.3.1 Firing pin protrusion. With the firing pin firmly seated in the base, each firing mechanism assembly shall be inspected for the required protrusion in both the drop-fire and lever-fire modes to the dimensional requirements of 3.4.1. Inspection shall be performed by gage listed in DL 7229669 which in turn is listed in Index DL 7238503. This examination shall be performed in conjunction with 4.6.4.2 Firing Mechanism Operation.

4.6.3.2 Material soundness. Each tube and base plug, biped, and baseplate shall be magnetic-particle inspected in accordance with applicable drawings and specifications. Material not conforming to the requirements of 3.3 and to applicable drawings and specifications shall be rejected.

4.6.3.3 Workmanship. Workmanship shall be inspected in accordance with MIL-C-13931.

4.6.3.4 Marking. Marking shall be exanined in accordance with MIL-C-13931.

4.6.4 Tests.

4.6.4.1 Mechanism function tests. The elevating, traversing and cant adjusting mechanism of each assembled mount shall be manually operated to determine the proper functioning of each sub-system. A minimum of two complete cycles of operation shall be performed on each of the three mechanisms. Any evidence of binding, scraping, rubbing, excessive play, or other malfunction of any component shall be cause for rejection.

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4.6.4.2 Firing mechanism operation. Each firing mechanism shall be inspected for smoothness of operation and freedom from interference, erratic movement and malfunction. The firing pin shall be fully retracted when the selector lever is in lever-fire position. See 4.6.3.1.

4.6.5 Performance test. The mortar assembly (cannon and mount) shall be subject to performance testing to assure compliance with 3.6 by a Government testing activity in accordance with procedures specified by the engineering and quality assurance elements of the procuring activity.

4.6.5.1 Functioning. Functioning shall be checked after testing in accordance with applicable acceptance test procedures to determine compliance with 3.6.1.

4.6.5.2 Bore. After testing, the bore shall be examined with a borescope and shall be measured with a star gage or other approved gage to determine compliance with 3.6.2.

4.6.5.3 Material soundness. After testing, the cannon tube base cap and housing plate, and the outer surfaces of the cannon tube shall be magnetic-particle tested in accordance with applicable drawings and specifications. Material not conforming to the requirements of 3.6.3 shall be rejected.

4.6.6 Inspection of preparation for delivery.

4.6.6.1 Packaging inspection. Unless otherwise specified, inspection to determine compliance with cleaning, preservation, packaging, packing and marking requirements of the applicable packaging data sheets for the level designated in the contract shall be performed in compliance with MIL-P-14232.

5. PREPARATION FOR DELIVERY.

5.1 First article package. On each contract, a first article package shall be provided in accordance with 3.1. The first article package for the mortar shall be unit packaged for the level of protection specified in the contract and packed level C. Unit packaging and packing shall be in accordance with MIL-P-14232 and Packaging Data Sheet: P 6732010. Marking shall be in accordance with MIL-STD-129.

5.2 Unit packaging packing and marking of the mortar. The mortar shall be unit packaged and picked in accordance with the requirements of MIL-P-14232 and Packaging Data Sheet P 6732010 for the level of protection specified in the contract (see 6.2).Marking shall be in accordance with MIL-STD-129.

5.3 Unit packaging, packing and marking of disassembled components and basic issue items. Disassembled components and basic issue items shall be unit packaged and packed in accordance with the requirements of MIL-P-14232 and Packaging Data Sheet P 6732010 for the level of protection specified in the contract (see 6.2). Marking shall be in accordance with MIL-STD-129.

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5.4 Unit packaging, packing and marking of repair parts. Repair parts shall be unit packaged and packed in accordance with the requirements of MIL-P-14232 and the applicable Packaging Data Sheets for the level of protection specified in the contract (see 6.2). Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cannon is intended for either lever-fire or drop-fire mode.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Requirements for submission of and shipping instructions for first article (see 3.1 and 4.4) and first article package (see 3.1 and 5.1).

(1) The contract should designate the examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

(2) The contract should specify the sample size (i.e., the number of units) of the first article.

(3) The contract should designate the inspection and test data that is required to be furnished the Government when the contractor is required to perform either or both examinations and tests.

(4) The contract should designate at whose expense a retest maybe performed.

(5) The contract should specify that the acceptable first article shall be delivered in accordance with the terms of the contract, or

(6) Any additional or extended examinations and tests beyond the scope of this specification should be specified and identified as a separate item of the contract.

c. List of basic issue items required.

d. Serial numbers to be used for the cannons and the replacement cannon assemblies.

e. List of inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.6.2.1).

f. Selection of applicable levels of preservation, packaging, and packing (see 5.2).

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g. Extent of supplier's responsibility for Government-furnished equipment.

h. Packaging instructions for basic issue items (see 5.3).

i. Applicable acceptance test procedures and frequency of testing (see 4.6.5).

6.2.2 Contract data requirements. Monthly reports of the results of final examination and performance testing shall be specified for delivery on DD Form 1423 in the contract.

6.3 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List, DD Form 1660.

6.4 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

6.5 Supersession data. This specification includes the requirements of Purchase Description WVTPD 1053 dated 28 August 1972.

Preparing activity:
Army-WV

Project Number 1010-A089

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-M-45965(WV) Mortar, 60MM: M19

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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